NeoSignal

AI Intelligence Daily · Multi-Source · Cross-Verified

22 May 2026

161 verified · 279 raw · 5 sources

161	4	157	0
TOTAL	VERIFIED	CONFIRMED	EMERGING

Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

VERIFIED	**0.80 - 1.00**	3+ independent sources including media
CONFIRMED	**0.50 - 0.79**	2 sources or 1 high-quality media
EMERGING	**0.25 - 0.49**	Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	142	Media
HackerNews	7	Community
TechCrunch AI	7	Media
The Verge AI	6	Media
MIT Tech Review	3	Media

VERIFIED INTELLIGENCE — 4 stories

VERI

Show HN: I Made a Claude Skill for Spec-Driven Development (SDD)

HackerNews +1 source 90% HN 26

At my work they provided a single Claude subscription for everyone on the team. To be honest I like kiro better as it provides a way better SDD management. But the company can't provide it and I can't afford it yet. Turns out I had the skill creator skill in my claude instance...

https://github.com/FredAntB/Spec-Driven-Development

VERI

Spotify is launching AI-generated remixes

The Verge AI +1 source 80%

Spotify and Universal Music Group (UMG) just announced a licensing deal that will allow users to prompt the creation of AI-generated remixes and covers for streaming songs. The tool will be a paid add-on for Premium subscribers. Artists will be able to opt out of the program,...

https://www.theverge.com/ai-artificial-intelligence/935379/spotify-umg-ai-covers-remix

VERI

Spotify Studio’s AI agent creates a daily podcast just for you

The Verge AI +1 source 80%

Studio by Spotify Labs is a new standalone AI app that generates a daily briefing, podcasts, and playlists on your PC using chatbot prompts. The AI-generated content draws from your Spotify listening history, as well as info from apps you connect to it, like your email inbox,...

https://www.theverge.com/entertainment/935390/spotify-studio-ai-app-personal-podcasts

VERI

Anthropic is paying $15 billion a year for access to Elon Musk’s data centers

The Verge AI +1 source 80%

Earlier this month, SpaceX and Anthropic announced a new compute partnership that provides access to the rocket company's Colossus data centers in Memphis, TN. Now, with the release of SpaceX's IPO filing, we have more details about that deal, including how much Anthropic is...

https://www.theverge.com/science/935229/spacex-anthropic-ipo-ai-capacity-deal-colossus

CONFIRMED SIGNALS — 157 stories

CONF

Show HN: Agent.email – sign up via curl, claim with a human OTP

HackerNews 50% HN 73

Hi HN! We're Haakam, Michael, and Adi from AgentMail- a ycs25 company. We give AI agents their own email inboxes. Recently, we ran an experiment called Agent.Email. It's a signup flow designed specifically for AI agents instead of humans. The inspiration came from a few comments...

https://news.ycombinator.com/item?id=48225596

CONF

Show HN: AI that interviews participants instead of holding another meeting

HackerNews 50% HN 8

No summary — see link for full article.

https://www.noada.app/

CONF

Show HN: Sylph – the open-source company brain behind my YC startup

HackerNews 50% HN 4

Hello HN! I'm Claire, founder of nao Labs (YC X25). Two months ago I started building a company brain in a git repo. Now I fully run my company with 8 AI agents, 20+ skills, and a self-learning context repo. I built it in a git repo because I refused to lock into any tool - not...

https://github.com/getnao/sylph

CONF

Show HN: AI Audiobook Narrator

HackerNews 50% HN 4

No summary — see link for full article.

https://warblize.com/

CONF

Show HN: Physics AI – Visualizing physics problems with step-by-step derivations

HackerNews 50% HN 3

No summary — see link for full article.

https://physicsai.chat

0 CONF

Show HN: My custom Statusline for Claude Code (Python wrapper around claudeline)

HackerNews 50% HN 3

Hey everyone - After lots of tweaking and tinkering to find what I like, sharing my favorite way to see my Claude Code statusline. Hope you like it too. The linked GH gist is a Python wrapper around claudeline that pipes its output through a few transformations and adds custom...

https://gist.github.com/Reebz/741c5647c860fe0b5214f39d9d887240

1 CONF

Trump delays AI security executive order, saying language ‘could have been a blocker’

TechCrunch AI 50%

President Trump delayed signing an executive order that would have required pre-release government security reviews of AI models, citing dissatisfaction with the order's language.

https://techcrunch.com/2026/05/21/trump-delays-ai-security-executive-order-i-dont-want-...

2 CONF

Hark raises $700M Series A for its secretive ‘universal’ AI interface

TechCrunch AI 50%

Hark expects to release its first multimodal models this summer, which it says will power a personal AI platform that works with existing products and services. The company expects to follow that with hardware devices built specifically for those systems.

https://techcrunch.com/2026/05/21/hark-raises-700m-series-a-for-its-secretive-universal...

3 CONF

Google is pitching an AI agent ecosystem to consumers who may not buy it

TechCrunch AI 50%

One of the most promising introductions at Google’s I/O developer conference on Tuesday was a new way for consumers to use the web: AI agents. Unfortunately, it was also the most confusing.

https://techcrunch.com/2026/05/21/google-is-pitching-an-ai-agent-ecosystem-to-consumers...

4 CONF

With aluminum prices up 20%, recycling startups bet on AI to cash in

TechCrunch AI 50%

Recycling startups are using AI to improve the recovery of critical minerals like aluminum, aiming to build a massive source of the metal.

https://techcrunch.com/2026/05/21/with-aluminum-prices-up-20-recycling-startups-bet-on-...

5 CONF

Roundtables: Can AI Learn to Understand the World?

MIT Tech Review **50%**

Listen to the session or watch below AI companies want to build systems that understand the external world and overcome the limitations of LLMs. Recent developments have brought world models to the forefront of the AI discussion. Watch a conversation with editor in chief Mat...

https://www.technologyreview.com/2026/05/21/1137756/roundtables-can-ai-learn-to-underst...

6 CONF

Scaling creativity in the age of AI

MIT Tech Review **50%**

Storytelling is core to humanity's DNA, stemming from our impulse to express ideals, warnings, hopes, and experiences. Technology has always been woven through the medium and the distribution: from early humans' innovation of natural pigments and charcoals for cave paintings to...

https://www.technologyreview.com/2026/05/21/1137613/scaling-creativity-in-the-age-of-ai/

7 CONF

Anthropic's Code with Claude showed off coding's future—whether you like it or not

MIT Tech Review **50%**

The vibes were strong at Code with Claude, Anthropic's two-day event for software developers in London that kicked off on May 19, the same day as Google's I/O in Palo Alto. (A coincidence, not a flex, Anthropic staffers assured me.) "Who here has shipped a pull request in the...

https://www.technologyreview.com/2026/05/21/1137735/anthropics-code-with-claude-showed-...

8 CONF

This AI guitar pedal let me roll my own effects

The Verge AI **50%**

I'm not sure anyone was really asking for an AI guitar pedal. But it was inevitable that someone would build one. One of the first to take the plunge is Polyend, a well-respected music gear maker with a reputation for building niche, idiosyncratic devices. The company has built...

https://www.theverge.com/ai-artificial-intelligence/935219/polyend-endless-ai-guitar-ef...

9 CONF

AI video is moving beyond clip slop

The Verge AI **50%**

This is Lowpass by Janko Roettgers, a newsletter on the ever-evolving intersection of tech and entertainment, syndicated just for The Verge subscribers once a week. Hollywood is cooked - or so a growing number of people on social media would like you to believe. Their purported...

https://www.theverge.com/column/935310/ai-video-luma-hollywood

0 CONF

Meta lays off thousands of employees to offset AI investments

The Verge AI **50%**

Meta has reportedly notified thousands of employees that they've been laid off as the company attempts to compensate for its hefty AI investments. In an email from Meta management shared by Business Insider, impacted staffers were told that the planned headcount reduction was...

https://www.theverge.com/tech/935163/meta-layoffs-ai-investment-offset-memo

1 **CONF**

SOLAR: A Self-Optimizing Open-Ended Autonomous Agent for Lifelong Learning and Continual Adaptation

ArXiv CS.AI 50%

arXiv:2605.20189v1 Announce Type: new Abstract: Despite the remarkable success of large language models (LLMs), they still face bottlenecks while deploying in dynamic, real-world settings with primary challenges being concept drift and the high cost of gradient-based adaptation....

https://arxiv.org/abs/2605.20189

2 **CONF**

High Quality Embeddings for Horn Logic Reasoning

ArXiv CS.AI 50%

arXiv:2605.20467v1 Announce Type: new Abstract: Neural networks can be trained to rank the choices made by logical reasoners, resulting in more efficient searches for answers. A key step in this process is creating useful embeddings, i.e., numeric representations of logical...

https://arxiv.org/abs/2605.20467

3 **CONF**

$ECUAS_n$: A family of metrics for principled evaluation of uncertainty-augmented systems

ArXiv CS.AI 50%

arXiv:2605.20490v2 Announce Type: new Abstract: In high-stakes automated decision-making, access to predictive uncertainty is essential for enabling users -- human or downstream systems -- to accept or reject predictions based on application-specific cost trade-offs. Such...

https://arxiv.org/abs/2605.20490

4 **CONF**

Open-World Evaluations for Measuring Frontier AI Capabilities

ArXiv CS.AI 50%

arXiv:2605.20520v1 Announce Type: new Abstract: Benchmark-based evaluation remains important for tracking frontier AI progress. But it can both overstate and understate deployed capability because it privileges tasks that can be precisely specified, automatically graded, easy to...

https://arxiv.org/abs/2605.20520

5 **CONF**

AgentAtlas: Beyond Outcome Leaderboards for LLM Agents

ArXiv CS.AI 50%

arXiv:2605.20530v1 Announce Type: new Abstract: Large language model agents now act on codebases, browsers, operating systems, calendars, files, and tool ecosystems, but the benchmarks used to evaluate them are fragmented: each emphasizes a different unit of measurement (final...

https://arxiv.org/abs/2605.20530

6 **CONF**

Personality Engineering with AI Agents: A New Methodology for Negotiation Research

ArXiv CS.AI 50%

arXiv:2605.20554v1 Announce Type: new Abstract: According to canonical negotiation theory, people's success in a negotiation depends on how well they balance competing demands--empathizing and asserting, demonstrating concern for other and concern for self, being soft on the...

https://arxiv.org/abs/2605.20554

7 **CONF**

Self-Improving Skill Learning for Robust Skill-based Meta-Reinforcement Learning

ArXiv CS.AI **50%**

arXiv:2502.03752v5 Announce Type: replace-cross Abstract: Meta-reinforcement learning (Meta-RL) facilitates rapid adaptation to unseen tasks but faces challenges in long-horizon environments. Skill-based approaches tackle this by decomposing state-action sequences into reusable...

https://arxiv.org/abs/2502.03752

8 **CONF**

Conditional Equivalence of DPO and RLHF: Implicit Assumption, Failure Modes, and Provable Alignment

ArXiv CS.AI **50%**

arXiv:2605.20834v1 Announce Type: new Abstract: Direct Preference Optimization (DPO) has emerged as a popular alternative to Reinforcement Learning from Human Feedback (RLHF), offering theoretical equivalence with simpler implementation. We prove this equivalence is conditional...

https://arxiv.org/abs/2605.20834

9 **CONF**

EvalMORAAL: Interpretable Chain-of-Thought and LLM-as-Judge Evaluation for Moral Alignment in Large Language Models

ArXiv CS.AI **50%**

arXiv:2510.05942v3 Announce Type: replace-cross Abstract: We present EvalMORAAL, a transparent chain-of-thought (CoT) framework that uses two scoring methods (log-probabilities and direct ratings) plus a model-as-judge peer review to evaluate moral alignment in 20 large language...

https://arxiv.org/abs/2510.05942

0 **CONF**

Playing Devil's Advocate: Off-the-Shelf Persona Vectors Rival Targeted Steering for Sycophancy

ArXiv CS.AI **50%**

arXiv:2605.21006v1 Announce Type: new Abstract: We study the effect of different persona on \textbf{sycophancy}: model's agreement with users even when the user is incorrect. The standard mitigation, Contrastive Activation Addition (CAA), derives a steering direction from...

https://arxiv.org/abs/2605.21006

1 **CONF**

Sutra: Tensor-Op RNNs as a Compilation Target for Vector Symbolic Architectures

ArXiv CS.AI **50%**

arXiv:2605.20919v1 Announce Type: cross Abstract: Sutra is a typed, purely functional programming language whose compiled forward pass is a PyTorch neural network. The compiler beta-reduces the whole program -- primitives, control flow, string I/O -- to one fused tensor-op graph...

https://arxiv.org/abs/2605.20919

2 **CONF**

Insights Generator: Systematic Corpus-Level Trace Diagnostics for LLM Agents

ArXiv CS.AI **50%**

arXiv:2605.21347v2 Announce Type: new Abstract: Diagnosing failures in LLM agents remains largely manual. Practitioners inspect a small subset of execution traces, form ad-hoc hypotheses, and iterate. This process misses patterns that only emerge across trace populations and...

https://arxiv.org/abs/2605.21347

3 CONF

Towards Resilient and Autonomous Networks: A BlueSky Vision on AI-Native 6G

ArXiv CS.AI 50%

arXiv:2605.21395v1 Announce Type: new Abstract: The proliferation of emerging applications, such as autonomous driving and immersive experiences, demands cellular networks that are not only faster, but fundamentally more resilient and autonomous. This paper presents a BlueSky...

https://arxiv.org/abs/2605.21395

4 CONF

Teaching AI Through Benchmark Construction: QuestBench as a Course-Based Practice for Accountable Knowledge Work

ArXiv CS.AI 50%

arXiv:2605.21413v2 Announce Type: new Abstract: As AI becomes part of everyday learning, many courses teach students to use it mainly as a productivity tool: how to prompt, search, summarize, write, code, and use tools more efficiently. We argue that AI education also needs a...

https://arxiv.org/abs/2605.21413

5 CONF

PALS: Power-Aware LLM Serving for Mixture-of-Experts Models

ArXiv CS.AI 50%

arXiv:2605.21427v1 Announce Type: new Abstract: Large language model (LLM) inference has become a dominant workload in modern data centers, driving significant GPU utilization and energy consumption. While prior systems optimize throughput and latency by batching, scheduling,...

https://arxiv.org/abs/2605.21427

6 CONF

AiraXiv: An AI-Driven Open-Access Platform for Human and AI Scientists

ArXiv CS.AI 50%

arXiv:2605.21481v1 Announce Type: new Abstract: Recent advances in artificial intelligence (AI) have accelerated the growth of both human-authored and AI-generated research outputs, placing increasing strain on traditional academic publishing systems and challenging the...

https://arxiv.org/abs/2605.21481

7 CONF

DeepWeb-Bench: A Deep Research Benchmark Demanding Massive Cross-Source Evidence and Long-Horizon Derivation

ArXiv CS.AI 50%

arXiv:2605.21482v1 Announce Type: new Abstract: Deep research, in which an agent searches the open web, collects evidence, and derives an answer through extended reasoning, is a prominent use case for frontier language models. Frontier deep research products score high on...

https://arxiv.org/abs/2605.21482

8 CONF

Q-Net: Queue Length Estimation via Kalman-based Neural Networks

ArXiv CS.AI 50%

arXiv:2509.24725v3 Announce Type: replace-cross Abstract: Estimating queue lengths at signalized intersections is a long-standing challenge in traffic management. Partial observability of vehicle flows complicates this task despite the availability of two privacy-preserving data...

https://arxiv.org/abs/2509.24725

9 **CONF**

GraphDiffMed: Knowledge-Constrained Differential Attention with Pharmacological Graph Priors for Medication Recommendation

ArXiv CS.AI 50%

arXiv:2605.20188v1 Announce Type: cross Abstract: Recommending safe and effective medication combinations from electronic health records (EHRs) is a core clinical AI problem, yet it remains difficult because patient trajectories are long, noisy, and clinically heterogeneous....

https://arxiv.org/abs/2605.20188

0 **CONF**

Parallel LLM Reasoning for Bias-Resilient, Robust Conceptual Abstraction

ArXiv CS.AI 50%

arXiv:2605.20194v1 Announce Type: cross Abstract: Large language models (LLMs) have been increasingly used to analyze text. However, they are often plagued with contextual reasoning limitations when analyzing long documents. When long documents are processed sequentially, early...

https://arxiv.org/abs/2605.20194

1 **CONF**

Data Scaling as Progressive Coverage of a Predictive Contribution Spectrum

ArXiv CS.AI 50%

arXiv:2605.20196v1 Announce Type: cross Abstract: We investigate the hypothesis that real-data scaling laws are governed by progressive coverage of a latent predictive contribution spectrum rather than by token-frequency tails alone. We work with a suffix-automaton...

https://arxiv.org/abs/2605.20196

2 **CONF**

Evaluating multimodal emotion recognition in proactive conversational agents: A user study

ArXiv CS.AI 50%

arXiv:2605.20200v1 Announce Type: cross Abstract: This article presents a multimodal emotion recognition module integrated into a proactive Socially Interactive Agent (SIA) powered by generative artificial intelligence. The system evaluates real-time affective states through two...

https://arxiv.org/abs/2605.20200

3 **CONF**

Long-Context Reasoning Through Proxy-Based Chain-of-Thought Tuning

ArXiv CS.AI 50%

arXiv:2605.20201v1 Announce Type: cross Abstract: Recent large language models support inputs of up to 10 million tokens, yet they perform poorly on long-context tasks that require complex reasoning. Such tasks can be solved using only a subset of the input -- a proxy context --...

https://arxiv.org/abs/2605.20201

4 **CONF**

ProcBench: Evaluating Process-Level Defects and Control Preservation in LLM Coding Agents

ArXiv CS.AI 50%

arXiv:2605.20251v2 Announce Type: cross Abstract: Existing benchmarks for LLM coding agents primarily evaluate final outcomes. While useful for measuring overall capability, these metrics provide limited visibility and often miss defects that arise during execution. We present...

https://arxiv.org/abs/2605.20251

5 CONF

GrandGuard: Taxonomy, Benchmark, and Safeguards for Elderly-Chatbot Interaction Safety

ArXiv CS.AI 50%

arXiv:2605.20203v1 Announce Type: cross Abstract: As older adults increasingly use LLM-based chatbots for companionship and assistance, a safety gap is emerging. Older adults may face vulnerabilities from social isolation, limited digital literacy, and cognitive decline, yet...

https://arxiv.org/abs/2605.20203

6 CONF

NeuroQA: A Large-Scale Image-Grounded Benchmark for 3D Brain MRI Understanding

ArXiv CS.AI 50%

arXiv:2605.20525v1 Announce Type: cross Abstract: We present NeuroQA, a large-scale benchmark for visual question answering in 3D brain magnetic resonance imaging (MRI), with 56,953 QA pairs from 12,977 subjects across 12 datasets. It spans ages 5-104 and five clinical domains:...

https://arxiv.org/abs/2605.20525

7 CONF

PrivacyAkinator: Articulating Key Privacy Design Decisions by Answering LLM-Generated Multiple-choice Questions

ArXiv CS.AI 50%

arXiv:2605.20206v1 Announce Type: cross Abstract: NIST's Privacy Risk Assessment Methodology (PRAM) provides a structured framework for privacy experts to assess privacy risks. However, its complexity and reliance on expert knowledge make it difficult for novice developers to...

https://arxiv.org/abs/2605.20206

8 CONF

TempGlitch: Evaluating Vision-Language Models for Temporal Glitch Detection in Gameplay Videos

ArXiv CS.AI 50%

arXiv:2605.21443v1 Announce Type: cross Abstract: Vision-language models (VLMs) are increasingly being explored for video game quality assurance, especially gameplay glitch detection. Most existing evaluations, however, treat glitches as static visual anomalies, asking models to...

https://arxiv.org/abs/2605.21443

9 CONF

AI-Assisted Competency Assessment from Egocentric Video in Simulation-Based Nursing Education

ArXiv CS.AI 50%

arXiv:2605.20233v1 Announce Type: cross Abstract: Assessing learner competency in clinical simulation requires expert observation that is time-intensive, difficult to scale, and subject to inter-rater variability. Vision-language models have emerged as a promising tool for...

https://arxiv.org/abs/2605.20233

0 CONF

Provably Learning Diffusion Models under the Manifold Hypothesis: Collapse and Refine

ArXiv CS.AI 50%

arXiv:2605.20235v1 Announce Type: cross Abstract: Diffusion models generate high-dimensional data with remarkable quality, yet how their training efficiently learns the score function, bypassing the curse of dimensionality when data is supported on low-dimensional manifolds,...

https://arxiv.org/abs/2605.20235

1 **CONF**

Efficient Table QA via TableGrid Navigation and Progressive Inference Prompting

ArXiv CS.AI 50%

arXiv:2605.20254v1 Announce Type: cross Abstract: Large Language Models (LLMs) have shown promising results on NLP tasks, however, their performance on tabular data still needs research attention, because Table Question-Answering (TQA) requires precise cell retrieval and...

https://arxiv.org/abs/2605.20254

2 **CONF**

Multi-Agent Reinforcement Learning for Safe Autonomous Driving Under Pedestrian Behavioral Uncertainty

ArXiv CS.AI 50%

arXiv:2605.20255v1 Announce Type: cross Abstract: Simulation-based testing of self-driving cars (SDCs) typically relies on scripted or simplified pedestrian models that do not capture the heterogeneity and uncertainty of real human crossing behavior. This limits the realism of...

https://arxiv.org/abs/2605.20255

3 **CONF**

FBOS-RL: Feedback-Driven Bi-Objective Synergistic Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2605.20256v1 Announce Type: cross Abstract: Reinforcement learning has become a cornerstone for aligning and unlocking the reasoning capabilities of large-scale models. At its core, the training loop of GRPO and its variants alternates between rollout sampling and policy...

https://arxiv.org/abs/2605.20256

4 **CONF**

APEX: Autonomous Policy Exploration for Self-Evolving LLM Agents

ArXiv CS.AI 50%

arXiv:2605.21240v1 Announce Type: cross Abstract: LLM agents have shown strong performance across a wide range of complex tasks, including interactive environments that require long-horizon decision making. But these agents cannot learn on the fly at test time. Self-evolving...

https://arxiv.org/abs/2605.21240

5 **CONF**

Generation of Heterogeneous PET Images from Uniform Organ Activity Maps Using a Pretrained Domain-Adapted Diffusion Model

ArXiv CS.AI 50%

arXiv:2605.20267v1 Announce Type: cross Abstract: Synthetic PET images are valuable for quantitative imaging workflow development, scalable virtual imaging trials, and deep learning model training, but conventional physics-based simulation approaches are computationally...

https://arxiv.org/abs/2605.20267

6 **CONF**

Chronicle: A Multimodal Foundation Model for Joint Language and Time Series Understanding

ArXiv CS.AI 50%

arXiv:2605.20268v1 Announce Type: cross Abstract: Real-world time series come with text: metadata, descriptions, news, reports. Yet time series foundation models process numerical sequences in isolation, and the multimodal text-and-time-series models that attempt to bridge the...

https://arxiv.org/abs/2605.20268

7 **CONF**

Conformal Selective Acting: Anytime-Valid Risk Control for RLVR-Trained LLMs

ArXiv CS.AI 50%

arXiv:2605.20270v1 Announce Type: cross Abstract: A local specialist LLM, fine-tuned with reinforcement learning from verifiable rewards (RLVR) on operator-local data, is installed in a regulated organization with per-deployment error budget α. The operator needs a safety...

https://arxiv.org/abs/2605.20270

8 **CONF**

JUDO: A Juxtaposed Domain-Oriented Multimodal Reasoner for Industrial Anomaly QA

ArXiv CS.AI 50%

arXiv:2605.20284v1 Announce Type: cross Abstract: Industrial anomaly detection has been significantly advanced by Large Multimodal Models (LMMs), enabling diverse human instructions beyond detection, particularly through visually grounded reasoning for better image...

https://arxiv.org/abs/2605.20284

9 **CONF**

Introspective X Training: Feedback Conditioning Improves Scaling Across all LLM Training Stages

ArXiv CS.AI 50%

arXiv:2605.20285v1 Announce Type: cross Abstract: We tackle the question of how to scale more efficiently across the many, ever-growing stages of current LLM training pipelines. Our guiding intuition stems from the fact that the dynamics of later stages of the pipeline, e.g....

https://arxiv.org/abs/2605.20285

0 **CONF**

Plug-and-Play Spiking Operators: Breaking the Nonlinearity Bottleneck in Spiking Transformers

ArXiv CS.AI 50%

arXiv:2605.20289v1 Announce Type: cross Abstract: ANN-to-SNN conversion offers a practical, training-free route to spiking large language models. However, current pipelines primarily focus on spike-driven realizations for Transformer linear-algebra operations, while providing...

https://arxiv.org/abs/2605.20289

1 **CONF**

Quant.npu: Enabling Efficient Mobile NPU Inference for on-device LLMs via Fully Static Quantization

ArXiv CS.AI 50%

arXiv:2605.20295v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly deployed on mobile devices, where Neural Processing Units (NPUs) necessitate fully static quantization for optimal inference efficiency. However, existing post-training quantization...

https://arxiv.org/abs/2605.20295

2 **CONF**

Robust Subspace-Constrained Quadratic Models for Low-Dimensional Structure Learning

ArXiv CS.AI 50%

arXiv:2605.20300v1 Announce Type: cross Abstract: In this paper, we propose a robust subspace-constrained quadratic model (SCQM) for learning low-dimensional structure from high-dimensional data. Building upon the subspace-constrained quadratic matrix factorization (SQMF)...

https://arxiv.org/abs/2605.20300

3 CONF

TASTE: A Designer-Annotated Multi-Dimensional Preference Dataset for AI-Generated Graphic Design

ArXiv CS.AI 50%

arXiv:2605.20731v1 Announce Type: cross Abstract: Text-to-image models produce graphic design at production scale, but their supervision comes from photo-style preference data with a single overall verdict per comparison. Designers evaluate along several distinct axes, including...

https://arxiv.org/abs/2605.20731

4 CONF

Less Data, Faster Training: repeating smaller datasets speeds up learning via sampling biases

ArXiv CS.AI 50%

arXiv:2605.20314v1 Announce Type: cross Abstract: This work investigates the ``small-vs-large gap'', where repeating on fewer samples can lead to compute saving during training compared to using a larger dataset. This is observed across algorithmic tasks, architectures and...

https://arxiv.org/abs/2605.20314

5 CONF

Representability-Aware Neural Networks for Reduced Density Matrices: Application to Fractional Chern Insulators

ArXiv CS.AI 50%

arXiv:2605.20326v1 Announce Type: cross Abstract: We develop a representability-aware and interpolable neural network (NN) framework for predicting two-particle reduced density matrices (2-RDMs). The NN incorporates a subset of representability conditions through its...

https://arxiv.org/abs/2605.20326

6 CONF

Security Document Classification with a Fine-Tuned Local Large Language Model: Benchmark Data and an Open-Source System

ArXiv CS.AI 50%

arXiv:2605.20368v1 Announce Type: cross Abstract: Organizations that scan documents for sensitive information face a practical problem. Cloud services require data to be sent to external infrastructure, while rule-based tools often miss threats that depend on context. This study...

https://arxiv.org/abs/2605.20368

7 CONF

SAVER: Selective As-Needed Vision Evidence for Multimodal Information Extraction

ArXiv CS.AI 50%

arXiv:2605.20713v1 Announce Type: cross Abstract: Multimodal IE in social media is difficult because a post may attach multiple images that are weakly related, redundant, or even misleading with respect to the text. In this setting, always-on multimodal fusion wastes computation...

https://arxiv.org/abs/2605.20713

8 CONF

Do as I Say, Not as I Do: Instruction-Induction Conflict in LLMs

ArXiv CS.AI 50%

arXiv:2605.20382v1 Announce Type: cross Abstract: Language models are trained to follow instructions, but they are also powerful pattern completers. What happens when these two objectives conflict? We construct conversations in which a user instruction to behave in a target way...

https://arxiv.org/abs/2605.20382

9 CONF

Decomposing MXFP4 quantization error for LLM reinforcement learning: reducible bias, recoverable deadzone, and an irreducible floor

ArXiv CS.AI 50%

arXiv:2605.20402v1 Announce Type: cross Abstract: MXFP4 arithmetic can dramatically accelerate reinforcement learning (RL) post-training of large language models (LLMs), yet the quantization error introduces severe accuracy degradation. Existing work treats the quantization...

https://arxiv.org/abs/2605.20402

0 CONF

Disentangling Sampling from Training Budget in Class-Imbalanced CT Body Composition Segmentation

ArXiv CS.AI 50%

arXiv:2605.20405v1 Announce Type: cross Abstract: Class imbalance is a fundamental challenge in medical image segmentation, where frequent classes typically dominate training at the expense of rare classes. Loss-based approaches mitigate imbalance by reweighting the per-pixel...

https://arxiv.org/abs/2605.20405

1 CONF

Mechanics of Bias and Reasoning: Interpreting the Impact of Chain-of-Thought Prompting on Gender Bias in LLMs

ArXiv CS.AI 50%

arXiv:2605.20410v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly deployed in socially sensitive settings despite substantial documentation that they encode gender biases. Chain-of-Thought (CoT) prompting has been proposed as a bias-mitigation...

https://arxiv.org/abs/2605.20410

2 CONF

Group-Algebraic Tensors: Provably-optimal Equivariant Learning and Physical Symmetry Discovery

ArXiv CS.AI 50%

arXiv:2605.20440v1 Announce Type: cross Abstract: We introduce the $\star_G$ tensor algebra, in which any finite group G defines the multiplication rule, making equivariance an intrinsic algebraic property rather than an architectural constraint. The framework rests on three...

https://arxiv.org/abs/2605.20440

3 CONF

Weight Decay Regimes in Grokking Transformers: Cheap Online Diagnostics

ArXiv CS.AI 50%

arXiv:2605.20441v1 Announce Type: cross Abstract: Transformers trained on modular arithmetic exhibit sharp transitions between memorization, generalization, and collapse. We show that weight decay acts as a scalar empirical control parameter for these regimes, and introduce two...

https://arxiv.org/abs/2605.20441

4 CONF

Comparative Evaluation of Deep Learning Models for Fake Image Detection

ArXiv CS.AI 50%

arXiv:2605.20971v1 Announce Type: cross Abstract: The growing sophistication of GAN-based image manipulation presents significant challenges for digital forensics. This study compares the performance of four pretrained CNN architectures including VGG16, ResNet50, EfficientNetB0,...

https://arxiv.org/abs/2605.20971

5 **CONF**

A Sharper Picture of Generalization in Transformers

ArXiv CS.AI **50%**

arXiv:2605.20988v1 Announce Type: cross Abstract: We study transformers' generalization behavior on boolean domains from the perspective of the Fourier Spectra of their target functions. In contrast to prior work (Edelman et al., 2022; Trauger and Tewari, 2024), which derived...

https://arxiv.org/abs/2605.20988

6 **CONF**

Variance Reduction for Expectations with Diffusion Teachers

ArXiv CS.AI **50%**

arXiv:2605.21489v1 Announce Type: cross Abstract: Pretrained diffusion models serve as frozen teachers feeding downstream pipelines such as text-to-3D, single-step distillation, and data attribution. The teacher gradients these pipelines consume are Monte Carlo (MC) expectations...

https://arxiv.org/abs/2605.21489

7 **CONF**

Tippett-minimum Fusion of Representation-space Diffusion Models for Multi-Encoder Out-of-Distribution Detection

ArXiv CS.AI **50%**

arXiv:2605.20502v1 Announce Type: cross Abstract: We address out-of-distribution (OOD) detection across the full spectrum of distribution shifts -- global domain changes, semantic divergence, texture differences, and covariate corruptions -- through a multi-encoder fusion of...

https://arxiv.org/abs/2605.20502

8 **CONF**

Codec-Robust Attacks on Audio LLMs

ArXiv CS.AI **50%**

arXiv:2605.20519v1 Announce Type: cross Abstract: Prior attacks on Audio Large Language Models (Audio LLMs) demonstrated that carefully crafted waveform-domain perturbations can force targeted adversarial outputs. As a defense mechanism against these attacks, real-world codec...

https://arxiv.org/abs/2605.20519

9 **CONF**

Machine-Learning-Enhanced Non-Invasive Testing for MASLD Fibrosis: Shallow-Deep Neural Networks Versus FIB-4, Tabular Foundation Models, and Large Language Models

ArXiv CS.AI **50%**

arXiv:2605.20523v1 Announce Type: cross Abstract: Advanced fibrosis is a major determinant of liver-related morbidity in metabolic dysfunction-associated steatotic liver disease (MASLD). FIB-4 is widely used as a first-line non-invasive test, but its fixed formula may underuse...

https://arxiv.org/abs/2605.20523

0 **CONF**

Collocational bootstrapping: A hypothesis about the learning of subject-verb agreement in humans and neural networks

ArXiv CS.AI **50%**

arXiv:2605.20529v1 Announce Type: cross Abstract: In what ways might statistical signals in linguistic input assist with the acquisition of syntax? Here we hypothesize a mechanism called collocational bootstrapping, in which regularities in word co-occurrence patterns can...

https://arxiv.org/abs/2605.20529

1 **CONF**

Axiomatizing Neural Networks via Pursuit of Subspaces

ArXiv CS.AI 50%

arXiv:2605.20534v1 Announce Type: cross Abstract: While deep neural networks have achieved remarkable success across a wide range of domains, their underlying mechanisms remain poorly understood, and they are often regarded as black boxes. This gap between empirical performance...

https://arxiv.org/abs/2605.20534

2 **CONF**

Self-Training Doesn't Flatten Language -- It Restructures It: Surface Markers Amplify While Deep Syntax Dies

ArXiv CS.AI 50%

arXiv:2605.20602v1 Announce Type: cross Abstract: Successive self-training on a language model's own outputs is widely characterized as a process of flattening: diversity drops, distributions narrow, and the text becomes "more like itself." We provide evidence that this...

https://arxiv.org/abs/2605.20602

3 **CONF**

Lower Bounds for Advection-Diffusion Equations: An Exploration with AI-Generated Proofs

ArXiv CS.AI 50%

arXiv:2605.20623v1 Announce Type: cross Abstract: We establish explicit lower bounds for advection-diffusion equations in three settings: a polynomial $\dot H^{-1}$ bound for inviscid shears with $u\in L^\infty_t W^{1,1}_y$, a uniform positive lower bound on the mixing scale for...

https://arxiv.org/abs/2605.20623

4 **CONF**

Accelerating Video Inverse Problem Solvers with Autoregressive Diffusion Models

ArXiv CS.AI 50%

arXiv:2605.20624v1 Announce Type: cross Abstract: Diffusion models provide powerful priors for zero-shot video inverse problems, but their real-time deployment is hindered by two inefficiencies: high initial latency caused by holistic video restoration, and low throughput...

https://arxiv.org/abs/2605.20624

5 **CONF**

Retrieval-Augmented Long-Context Translation for Cultural Image Captioning: Gators submission for AmericasNLP 2026 shared task

ArXiv CS.AI 50%

arXiv:2605.20626v1 Announce Type: cross Abstract: We present the University of Florida Gators submission to the AmericasNLP 2026 shared task on cultural image captioning for Indigenous languages. Our two-stage pipeline generates a Spanish intermediate caption with Qwen2.5-VL,...

https://arxiv.org/abs/2605.20626

6 **CONF**

Pareto-Enhanced Portrait Generation: Vision-Aligned Text Supervision for Alignment, Realism, and Aesthetics

ArXiv CS.AI 50%

arXiv:2605.20640v1 Announce Type: cross Abstract: Text-to-image diffusion models often face a severe trilemma in human portrait generation: text-image alignment, photorealism, and human-perceived aesthetics inherently inhibit one another. Supervised Fine-Tuning (SFT) is an...

https://arxiv.org/abs/2605.20640

7 CONF

Trusted Weights, Treacherous Optimizations? Optimization-Triggered Backdoor Attacks on LLMs

ArXiv CS.AI 50%

arXiv:2605.20641v1 Announce Type: cross Abstract: Inference optimization is a vital technique for deploying LLMs at scale. Compilation is the most widely adopted optimization technique for LLMs. While it assumes semantic equivalence between the original and compiled graphs, we...

https://arxiv.org/abs/2605.20641

8 CONF

Design for Manufacturing: A Manufacturability Knowledge-Integrated Reinforcement Learning Framework for Free-Form Pipe Routing in Aeroengines

ArXiv CS.AI 50%

arXiv:2605.20644v1 Announce Type: cross Abstract: Design for manufacturing plays a critical role in advanced aeroengine development, where complex components necessitate careful consideration of manufacturability. However, current practices in pipe routing remain largely...

https://arxiv.org/abs/2605.20644

9 CONF

REFLECTOR: Internalizing Step-wise Reflection against Indirect Jailbreak

ArXiv CS.AI 50%

arXiv:2605.20654v1 Announce Type: cross Abstract: While Large Language Models (LLMs) demonstrate remarkable capabilities, they remain susceptible to sophisticated, multi-step jailbreak attacks that circumvent conventional surface-level safety alignment by exploiting the internal...

https://arxiv.org/abs/2605.20654

0 CONF

On the limits and opportunities of AI reviewers: Reviewing the reviews of Nature-family papers with 45 expert scientists

ArXiv CS.AI 50%

arXiv:2605.20668v1 Announce Type: cross Abstract: With the advancement of AI capabilities, AI reviewers are beginning to be deployed in scientific peer review, yet their capability and credibility remain in question: many scientists simply view them as probabilistic systems...

https://arxiv.org/abs/2605.20668

1 CONF

DIVE: Embedding Compression via Self-Limiting Gradient Updates

ArXiv CS.AI 50%

arXiv:2605.20689v1 Announce Type: cross Abstract: High-dimensional embeddings from large language models impose significant storage and computational costs on vector search systems. Recent embedding compression methods, including Matryoshka-Adaptor (EMNLP 2024), Search-Adaptor...

https://arxiv.org/abs/2605.20689

2 CONF

Heartbeat-Bound Hierarchical Credentials: Cryptographic Revocation for AI Agent Swarms

ArXiv CS.AI 50%

arXiv:2605.20704v1 Announce Type: cross Abstract: Autonomous AI agents that spawn sub-agent swarms create a safety gap: existing credential revocation mechanisms, OAuth~2.0 introspection, OCSP, and W3C Status Lists, require network connectivity to a central authority, leaving...

https://arxiv.org/abs/2605.20704

3 **CONF**

Stdlib or Third-Party? Empirical Performance and Correctness of LLM-Assisted Zero-Dependency Python Libraries

ArXiv CS.AI 50%

arXiv:2605.21405v1 Announce Type: cross Abstract: Third-party Python libraries introduce dependency management overhead, supply chain risk, and deployment friction in constrained environments. A natural question is how much of this ecosystem can be replicated using only Python's...

https://arxiv.org/abs/2605.21405

4 **CONF**

DeCoR: Design and Control Co-Optimization for Urban Streets Using Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2605.21311v1 Announce Type: cross Abstract: Modern vision systems can detect, track, and forecast urban actors at scale, yet translating perception outputs to urban design remains limited. We introduce DeCoR, a two-stage reinforcement learning framework that leverages flow...

https://arxiv.org/abs/2605.21311

5 **CONF**

An Application-Layer Multi-Modal Covert-Channel Reference Monitor for LLM Agent Egress

ArXiv CS.AI 50%

arXiv:2605.20734v1 Announce Type: cross Abstract: A large language model (LLM) agent that sends messages can leak data inside them. Destination allowlists and content scanners do not police whether an otherwise-benign payload is itself a covert channel: a compromised agent...

https://arxiv.org/abs/2605.20734

6 **CONF**

Closed Loop Dynamic Driving Data Mixture for Real-Synthetic Co-Training

ArXiv CS.AI 50%

arXiv:2605.21372v1 Announce Type: cross Abstract: Data scaling is fundamental to modern deep learning, and grows increasingly critical as autonomous driving shifts to end-to-end learning. Real-world driving data is expensive to annotate and scene-biased, making real-synthetic...

https://arxiv.org/abs/2605.21372

7 **CONF**

GraphRAG on Consumer Hardware: Benchmarking Local LLMs for Healthcare EHR Schema Retrieval

ArXiv CS.AI 50%

arXiv:2605.20815v1 Announce Type: cross Abstract: Graph-based Retrieval Augmented Generation (GraphRAG) extends retrieval-augmented generation to support structured reasoning over complex corpora, but its reliability under resource-constrained, privacy-sensitive deployments...

https://arxiv.org/abs/2605.20815

8 **CONF**

GenAI-Driven Threat Detection with Microsoft Security Copilot

ArXiv CS.AI 50%

arXiv:2605.20896v1 Announce Type: cross Abstract: Defending against today's increasingly sophisticated cyberattacks requires security analysts to continuously translate evolving attacker tradecraft into detection logic. This places defenders in a reactive posture, requiring...

https://arxiv.org/abs/2605.20896

9 **CONF**

Comparing Explanations is Not Enough, Explain the Change: New Standards are Needed to Explain Behavioral Shifts in Large Language Models

ArXiv CS.AI **50%**

arXiv:2602.02304v2 Announce Type: replace Abstract: Large-scale foundation models exhibit \emph{behavioral shifts} when subjected to interventions such as scaling, fine-tuning, reinforcement learning with human feedback, or in-context learning. Current explainability methods are...

https://arxiv.org/abs/2602.02304

00 **CONF**

Winfree Oscillatory Neural Network

ArXiv CS.AI **50%**

arXiv:2605.20922v1 Announce Type: cross Abstract: Oscillations and synchronization are widely believed to play a fundamental role in representation and computation. However, existing machine learning approaches based on synchronization dynamics have largely been confined to...

https://arxiv.org/abs/2605.20922

01 **CONF**

Causal Past Logic for Runtime Verification of Distributed LLM Agent Workflows

ArXiv CS.AI **50%**

arXiv:2605.20923v1 Announce Type: cross Abstract: Distributed LLM agent workflows should not be monitored as if they produced a single sequential log. In an asynchronous execution, a decision can only depend on events that are causally visible to the lifeline that makes it: an...

https://arxiv.org/abs/2605.20923

02 **CONF**

Beyond Attention Scores: SVD-Based Vision Token Pruning for Efficient Vision-Language Models

ArXiv CS.AI **50%**

arXiv:2604.11530v2 Announce Type: replace-cross Abstract: Vision-Language Models (VLMs) have revolutionized multi-modal learning by jointly processing visual and textual information. Yet, they face significant challenges due to the high computational and memory demands of...

https://arxiv.org/abs/2604.11530

03 **CONF**

Towards Context-Invariant Safety Alignment for Large Language Models

ArXiv CS.AI **50%**

arXiv:2605.20994v1 Announce Type: cross Abstract: Preference-based post-training aligns LLMs with human intent, yet safety behavior often remains brittle. A model may refuse a harmful request in a standard prompt but comply when the same intent is wrapped in adversarial wording....

https://arxiv.org/abs/2605.20994

04 **CONF**

Hybrid Machine Learning Model for Forest Height Estimation from TanDEM-X and Landsat Data

ArXiv CS.AI **50%**

arXiv:2605.20997v1 Announce Type: cross Abstract: Integrating machine learning (ML) with physical models (PM) has emerged as a promising way of retrieving geophysical parameters from remote sensing data. In this context, a ML model for estimating forest height from TanDEM-X...

https://arxiv.org/abs/2605.20997

05 **CONF**

Beyond Text-to-SQL: An Agentic LLM System for Governed Enterprise Analytics APIs

ArXiv CS.AI **50%**

arXiv:2605.21027v1 Announce Type: cross Abstract: Enterprise analytics aims to make organizational data accessible for decision-making, yet non-technical users still face barriers when using traditional business intelligence tools or Text-to-SQL systems. While recent Text-to-SQL...

https://arxiv.org/abs/2605.21027

06 **CONF**

Divide et Calibra: Multiclass Local Calibration via Vector Quantization

ArXiv CS.AI **50%**

arXiv:2605.21060v1 Announce Type: cross Abstract: Accurate and well-calibrated Machine Learning (ML) models are mandatory in high-stakes settings, yet effective multiclass calibration remains challenging: global approaches assume calibration errors are homogeneous across the...

https://arxiv.org/abs/2605.21060

07 **CONF**

Fine-grained Claim-level RAG Benchmark for Law

ArXiv CS.AI **50%**

arXiv:2605.21071v2 Announce Type: cross Abstract: The rapid progress of large language models (LLMs) is shifting semantic search toward a question-answering paradigm, where users ask questions and LLMs generate responses. In high-stake domains such as law, retrieval-augmented...

https://arxiv.org/abs/2605.21071

08 **CONF**

Decoupling Communication from Policy: Robust MARL under Bandwidth Constraints

ArXiv CS.AI **50%**

arXiv:2605.21085v1 Announce Type: cross Abstract: Communication enables coordination in multi-agent reinforcement learning (MARL), but many real-world applications, e.g., search-and-rescue with drone swarms, operate under severe bandwidth constraints. Many communication...

https://arxiv.org/abs/2605.21085

09 **CONF**

On the Complexity of Entailment for Cumulative Propositional Dependence Logics

ArXiv CS.AI **50%**

arXiv:2605.21113v1 Announce Type: cross Abstract: This paper establishes and proves complexity results for entailment for cumulative propositional dependence logic and for cumulative propositional logic with team semantics. As recently shown, cumulative logics are famously...

https://arxiv.org/abs/2605.21113

10 **CONF**

SAM-Sode: Towards Faithful Explanations for Tiny Bacteria Detection

ArXiv CS.AI **50%**

arXiv:2605.21186v1 Announce Type: cross Abstract: Interpretability in object detection provides crucial confidence support for clinical auxiliary diagnosis. However, in tiny bacteria detection, traditional explanation methods often suffer from blurred foreground boundaries and...

https://arxiv.org/abs/2605.21186

11 **CONF**

SURGE: An Event-Centric Social Media Sentiment Time Series Benchmark with Interaction Structure

ArXiv CS.AI 50%

arXiv:2605.21198v1 Announce Type: cross Abstract: Public events on social media generate large volumes of discussion whose collective dynamics carry direct value for opinion forecasting and crisis response. Capturing how these dynamics evolve across an event's lifecycle requires...

https://arxiv.org/abs/2605.21198

12 **CONF**

Enhanced Reinforcement Learning-based Process Synthesis via Quantum Computing

ArXiv CS.AI 50%

arXiv:2605.21213v1 Announce Type: cross Abstract: In this work, we present quantum reinforcement learning (RL) as a solution strategy for process synthesis problems. Building on our prior work, we develop a generalized framework that formally poses process synthesis as a Markov...

https://arxiv.org/abs/2605.21213

13 **CONF**

Artificial Intelligence Reshapes Microwave Photonics

ArXiv CS.AI 50%

arXiv:2605.21224v1 Announce Type: cross Abstract: As a rapidly emerging interdisciplinary field that intrinsically integrates microwave and photonics, microwave photonics (MWP) provides disruptive solutions to overcome the fundamental bandwidth of conventional electronic...

https://arxiv.org/abs/2605.21224

14 **CONF**

PREFINE: Preference-Based Implicit Reward and Cost Fine-Tuning for Safety Alignment

ArXiv CS.AI 50%

arXiv:2605.21225v1 Announce Type: cross Abstract: We address the problem of making a pre-trained reinforcement learning (RL) policy safety-aware by incorporating cost constraints without retraining it from scratch. While costs could be numerically encoded, we assume a more...

https://arxiv.org/abs/2605.21225

15 **CONF**

OCTOPUS: Optimized KV Cache for Transformers via Octahedral Parametrization Under optimal Squared error quantization

ArXiv CS.AI 50%

arXiv:2605.21226v1 Announce Type: cross Abstract: The key-value (KV) cache dominates memory bandwidth and footprint in long-context autoregressive inference. Recent rotation-preconditioned codecs (TurboQuant, PolarQuant) show that a structured random rotation followed by a...

https://arxiv.org/abs/2605.21226

16 **CONF**

Stochastic MeanFlow Policies: One-Step Generative Control with Entropic Mirror Descent

ArXiv CS.AI 50%

arXiv:2605.21282v2 Announce Type: cross Abstract: Online off-policy reinforcement learning (RL) is shaped by two coupled choices: the policy class and the update rule. Gaussian policies are fast and have tractable entropy, but struggle with multimodal action distributions....

https://arxiv.org/abs/2605.21282

17 **CONF**

Do LLMs Triage Like Clinicians? A Dynamic Study of Outpatient Referral

ArXiv CS.AI **50%**

arXiv:2503.08292v5 Announce Type: replace-cross Abstract: Outpatient referral (OR) is a core clinical workflow that assigns patients to hospital departments under incomplete and evolving information, yet it is commonly simplified as a static classification problem despite being...

https://arxiv.org/abs/2503.08292

18 **CONF**

TimeSRL: Generalizable Time-Series Behavioral Modeling via Semantic RL-Tuned LLMs -- A Case Study in Mental Health

ArXiv CS.AI **50%**

arXiv:2605.21295v1 Announce Type: cross Abstract: Longitudinal passive sensing enables continuous health prediction, yet models often fail under cross-dataset distribution shifts. Traditional ML overfits cohort-specific artifacts, while Large Language Models (LLMs) struggle to...

https://arxiv.org/abs/2605.21295

19 **CONF**

Frontier: Towards Comprehensive and Accurate LLM Inference Simulation

ArXiv CS.AI **50%**

arXiv:2605.21312v1 Announce Type: cross Abstract: Modern LLM serving is no longer homogeneous or monolithic. Production systems now combine disaggregated execution, complex parallelism, runtime optimizations, and stateful workloads such as reasoning, agents, and RL rollouts....

https://arxiv.org/abs/2605.21312

20 **CONF**

SymbolicLight V1: Spike-Gated Dual-Path Language Modeling with High Activation Sparsity and Sub-Billion-Scale Pre-Training Evidence

ArXiv CS.AI **50%**

arXiv:2605.21333v1 Announce Type: cross Abstract: Natively trained spiking language models struggle to combine Transformer-like language quality, stable multi-domain pre-training, and high activation sparsity. We present SymbolicLight V1, a spike-gated dual-path language model...

https://arxiv.org/abs/2605.21333

21 **CONF**

How to Build Marcus's Algebraic Mind: Algebro-Deterministic Substrate over Galois Fields

ArXiv CS.AI **50%**

arXiv:2605.21379v2 Announce Type: cross Abstract: In The Algebraic Mind, Gary Marcus identified three components essential for any adequate cognitive architecture: operations over variables, recursively structured representations, and a distinction between mental representations...

https://arxiv.org/abs/2605.21379

22 **CONF**

On the Regularity and Generalization of One-Step Wasserstein-guided Generative Models for PDE-Induced Measures

ArXiv CS.AI **50%**

arXiv:2605.21388v1 Announce Type: cross Abstract: Despite the remarkable empirical success of generative models, the available theory on their statistical accuracy in scientific computing remains largely pessimistic. This paper develops a theoretical framework for understanding...

https://arxiv.org/abs/2605.21388

23 CONF

Designing Conversations with the Dead: How People Engage with Generative Ghosts

ArXiv CS.AI 50%

arXiv:2605.21390v1 Announce Type: cross Abstract: We examine how people experience two choices in the design of generative ghosts, AI systems that are trained on data of the dead: representation, where an AI speaks about a deceased person in the third person, and reincarnation,...

https://arxiv.org/abs/2605.21390

24 CONF

Open-source LLMs administer maximum electric shocks in a Milgram-like obedience experiment

ArXiv CS.AI 50%

arXiv:2605.21401v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly deployed as autonomous agents that make sequences of decisions over extended interactions in high-stakes domains. However, the behavior of LLMs under sustained authority pressure is...

https://arxiv.org/abs/2605.21401

25 CONF

torchtune: PyTorch native post-training library

ArXiv CS.AI 50%

arXiv:2605.21442v1 Announce Type: cross Abstract: Modern LLMs typically require multistage training pipelines to achieve strong downstream performance, with post-training serving as the main interface for adapting open-weight models. We introduce torchtune, a PyTorch-native...

https://arxiv.org/abs/2605.21442

26 CONF

HITL-D: Human In The Loop Diffusion Assisted Shared Control

ArXiv CS.AI 50%

arXiv:2605.21460v1 Announce Type: cross Abstract: Autonomous manipulation systems have achieved remarkable capabilities, yet the integration of human expertise with diffusion-based policies in shared control remains relatively unexplored. In this paper, we propose...

https://arxiv.org/abs/2605.21460

27 CONF

WikiVQABench: A Knowledge-Grounded Visual Question Answering Benchmark from Wikipedia and Wikidata

ArXiv CS.AI 50%

arXiv:2605.21479v1 Announce Type: cross Abstract: Visual Question Answering (VQA) benchmarks have largely emphasized perception-based tasks that can be solved from visual content alone. In contrast, many real-world scenarios require external knowledge that is not directly...

https://arxiv.org/abs/2605.21479

28 CONF

Quantifying Hyperparameter Transfer and the Importance of Embedding Layer Learning Rate

ArXiv CS.AI 50%

arXiv:2605.21486v1 Announce Type: cross Abstract: Hyperparameter transfer allows extrapolating optimal optimization hyperparameters from small to large scales, making it critical for training large language models (LLMs). This is done either by fitting a scaling law to the...

https://arxiv.org/abs/2605.21486

29 CONF

Universal Reasoner: A Single, Composable Plug-and-Play Reasoner for Frozen LLMs

ArXiv CS.AI 50%

arXiv:2505.19075v3 Announce Type: replace Abstract: Large Language Models (LLMs) have demonstrated remarkable general capabilities, but enhancing skills such as reasoning often demands substantial computational resources and may compromise generalization. While...

https://arxiv.org/abs/2505.19075

30 CONF

DeFacto: Counterfactual Thinking with Images for Enforcing Evidence-Grounded and Faithful Reasoning

ArXiv CS.AI 50%

arXiv:2509.20912v4 Announce Type: replace Abstract: Recent advances in multimodal language models (MLLMs) have made thinking with images a dominant paradigm for multimodal reasoning. However, existing methods still fail to ensure evidence-answer consistency, where correct...

https://arxiv.org/abs/2509.20912

31 CONF

Agentic Physical AI toward a Domain-Specific Foundation Model for Nuclear Reactor Control

ArXiv CS.AI 50%

arXiv:2512.23292v3 Announce Type: replace Abstract: The prevailing paradigm in AI for physical systems (scaling general-purpose foundation models toward universal multimodal reasoning) confronts a fundamental barrier at the control interface. Recent benchmarks show that even...

https://arxiv.org/abs/2512.23292

32 CONF

Chain-of-thought obfuscation learned from output supervision can generalise to unseen tasks

ArXiv CS.AI 50%

arXiv:2601.23086v2 Announce Type: replace Abstract: Chain-of-thought (CoT) reasoning provides a significant performance uplift to LLMs by enabling planning, exploration, and deliberation of their actions. CoT is also a powerful tool for monitoring the behaviours of these agents:...

https://arxiv.org/abs/2601.23086

33 CONF

MARS: Modular Agent with Reflective Search for Automated AI Research

ArXiv CS.AI 50%

arXiv:2602.02660v3 Announce Type: replace Abstract: A critical bottleneck in automating AI research is the execution of complex machine learning engineering (MLE) tasks. MLE differs from general software engineering due to computationally expensive evaluation (e.g., model...

https://arxiv.org/abs/2602.02660

34 CONF

Learning to Configure Agentic AI Systems

ArXiv CS.AI 50%

arXiv:2602.11574v3 Announce Type: replace Abstract: Configuring LLM-based agent systems involves choosing workflows, tools, token budgets, and prompts from a large combinatorial design space, and is typically handled today by fixed templates or hand-tuned heuristics that apply...

https://arxiv.org/abs/2602.11574

35 CONF

Retaining Suboptimal Actions to Follow Shifting Optima in Multi-Agent Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2602.17062v2 Announce Type: replace Abstract: Value decomposition is a core approach for cooperative multi-agent reinforcement learning (MARL). However, existing methods still rely on a single optimal action and struggle to adapt when the underlying value function shifts...

https://arxiv.org/abs/2602.17062

36 CONF

FT-Dojo: Towards Autonomous LLM Fine-Tuning with Language Agents

ArXiv CS.AI 50%

arXiv:2603.01712v2 Announce Type: replace Abstract: Fine-tuning large language models for vertical domains remains labor-intensive, requiring practitioners to curate data, configure training, and iteratively diagnose model behavior. Despite growing interest in autonomous machine...

https://arxiv.org/abs/2603.01712

37 CONF

LLM Jaggedness Unlocks Scientific Creativity

ArXiv CS.AI 50%

arXiv:2605.10574v2 Announce Type: replace Abstract: As artificial intelligence advances, models are not improving uniformly. Instead, progress unfolds in a jagged fashion, with capabilities growing unevenly across tasks, domains, and model scales. In this work, we examine this...

https://arxiv.org/abs/2605.10574

38 CONF

ComplexMCP: Evaluation of LLM Agents in Dynamic, Interdependent, and Large-Scale Tool Sandbox

ArXiv CS.AI 50%

arXiv:2605.10787v2 Announce Type: replace Abstract: Current LLM agents are proficient at calling isolated APIs but struggle with the "last mile" of commercial software automation. In real-world scenarios, tools are not independent; they are atomic, interdependent, and prone to...

https://arxiv.org/abs/2605.10787

39 CONF

SVFSearch: A Multimodal Knowledge-Intensive Benchmark for Short-Video Frame Search in the Gaming Vertical Domain

ArXiv CS.AI 50%

arXiv:2605.17946v2 Announce Type: replace Abstract: Multimodal large language models are increasingly used as agent backbones that understand multimodal inputs, plan retrieval actions, invoke external tools, and reason over retrieved information. Yet existing benchmarks rarely...

https://arxiv.org/abs/2605.17946

40 CONF

CHEM: Estimating and Understanding Hallucinations in Deep Learning for Image Processing

ArXiv CS.AI 50%

arXiv:2512.09806v2 Announce Type: replace-cross Abstract: Deep learning-based methods have recently achieved significant success in image reconstruction problems. However, challenges have emerged, as these methods may generate unrealistic artifacts or hallucinations, which can...

https://arxiv.org/abs/2512.09806

41 CONF

LLMs on the Line: Data Determines Loss-to-Loss Scaling Laws

ArXiv CS.AI 50%

arXiv:2502.12120v3 Announce Type: replace-cross Abstract: Scaling laws guide the development of large language models (LLMs) by offering estimates for the optimal balance of model size, tokens, and compute. More recently, loss-to-loss scaling laws that relate losses across...

https://arxiv.org/abs/2502.12120

42 CONF

How Many Human Survey Respondents is a Large Language Model Worth? An Uncertainty Quantification Perspective

ArXiv CS.AI 50%

arXiv:2502.17773v5 Announce Type: replace-cross Abstract: Large language models (LLMs) are increasingly used to simulate survey responses, but synthetic data can be misaligned with the human population, leading to unreliable inference. We develop a general framework that...

https://arxiv.org/abs/2502.17773

43 CONF

Beyond Words: Multimodal LLM Knows When to Speak

ArXiv CS.AI 50%

arXiv:2505.14654v2 Announce Type: replace-cross Abstract: Chatbots via large language models (LLMs) generate fluent responses but often struggle with when to speak, especially for brief, timely listener reactions during ongoing dialogue. We present a multimodal strategy for...

https://arxiv.org/abs/2505.14654

44 CONF

Time-Prompt: Integrated Heterogeneous Prompts for Unlocking LLMs in Time Series Forecasting

ArXiv CS.AI 50%

arXiv:2506.17631v4 Announce Type: replace-cross Abstract: Time series forecasting aims to model temporal dependencies among variables for future state inference, holding significant importance and widespread applications in real-world scenarios. Although deep learning-based...

https://arxiv.org/abs/2506.17631

45 CONF

M3: Conversational LLMs Simplify Secure Clinical Data Access, Understanding, and Analysis

ArXiv CS.AI 50%

arXiv:2507.01053v4 Announce Type: replace-cross Abstract: Large-scale clinical databases offer opportunities for medical research, but their complexity creates barriers to effective use. The Medical Information Mart for Intensive Care (MIMIC-IV), one of the world's largest...

https://arxiv.org/abs/2507.01053

46 CONF

FAIR-Pruner: A Flexible Framework for Automatic Layer-Wise Pruning via Tolerance of Difference

ArXiv CS.AI 50%

arXiv:2508.02291v3 Announce Type: replace-cross Abstract: Structured pruning is a standard tool for compressing deep neural networks, but its practical performance depends on how sparsity is allocated across layers. We propose FAIR-Pruner, a search-free framework for adaptive...

https://arxiv.org/abs/2508.02291

47 CONF

Measuring and mitigating overreliance to build human-compatible AI

ArXiv CS.AI 50%

arXiv:2509.08010v2 Announce Type: replace-cross Abstract: Large language models (LLMs) distinguish themselves from previous technologies by functioning as collaborative ``thought partners,'' capable of engaging more fluidly in natural language on a range of tasks. As LLMs...

https://arxiv.org/abs/2509.08010

48 CONF

CTFExplorer: Evaluating LLM Offensive Agents Through Multi-Target Web CTF Benchmarking

ArXiv CS.AI 50%

arXiv:2602.08023v3 Announce Type: replace-cross Abstract: Existing benchmarks for LLM-based offensive security agents use isolated, single-target setups with a known vulnerable service and fixed objective. They measure exploitation effectively, but miss how real Capture-the-Flag...

https://arxiv.org/abs/2602.08023

49 CONF

DrugRAG: Enhancing Pharmacy LLM Performance Through A Novel Retrieval-Augmented Generation Pipeline

ArXiv CS.AI 50%

arXiv:2512.14896v2 Announce Type: replace-cross Abstract: In our study, we evaluated large language model (LLM) performance on pharmacy licensure-style question-answering tasks and developed an external knowledge integration method to improve accuracy. We benchmarked ten LLMs...

https://arxiv.org/abs/2512.14896

50 CONF

Mind the Generative Details: Direct Localized Detail Preference Optimization for Video Diffusion Models

ArXiv CS.AI 50%

arXiv:2601.04068v4 Announce Type: replace-cross Abstract: Aligning text-to-video diffusion models with human preferences is crucial for generating high-quality videos. Existing Direct Preference Otimization (DPO) methods rely on multi-sample ranking and task-specific critic...

https://arxiv.org/abs/2601.04068

51 CONF

MCP-Atlas: A Large-Scale Benchmark for Tool-Use Competency with Real MCP Servers

ArXiv CS.AI 50%

arXiv:2602.00933v3 Announce Type: replace-cross Abstract: The Model Context Protocol (MCP) is emerging as a standard interface through which large language model (LLM) agents discover and invoke external tools. However, existing MCP evaluations fall short along three key axes:...

https://arxiv.org/abs/2602.00933

52 CONF

Multimodal Optimal Transport for Training-free Temporal Segmentation in Surgical Robotics

ArXiv CS.AI 50%

arXiv:2602.24138v2 Announce Type: replace-cross Abstract: Automated recognition of surgical phases and steps is a fundamental capability for intraoperative decision support, workflow automation, and skill assessment in robotic-assisted surgery. Existing approaches either depend...

https://arxiv.org/abs/2602.24138

53 CONF

Iterative LLM-based improvement for French Clinical Interview Transcription and Speaker Diarization

ArXiv CS.AI 50%

arXiv:2603.00086v2 Announce Type: replace-cross Abstract: Automatic speech recognition for French medical conversations remains challenging, with word error rates often exceeding 30% in spontaneous clinical speech. This study proposes a multi-pass LLM post-processing...

https://arxiv.org/abs/2603.00086

54 CONF

MASFactory: A Graph-centric Framework for Orchestrating LLM-Based Multi-Agent Systems with Vibe Graphing

ArXiv CS.AI 50%

arXiv:2603.06007v2 Announce Type: replace-cross Abstract: Large language model-based (LLM-based) multi-agent systems (MAS) are increasingly used to extend agentic problem solving via role specialization and collaboration. MAS workflows can be naturally modeled as directed...

https://arxiv.org/abs/2603.06007

55 CONF

Deeper Thought, Weaker Aim: Understanding and Mitigating Perceptual Impairment during Reasoning in Multimodal Large Language Models

ArXiv CS.AI 50%

arXiv:2603.14184v2 Announce Type: replace-cross Abstract: Multimodal large language models (MLLMs) often suffer from perceptual impairments under extended reasoning modes, particularly in visual question answering (VQA) tasks. We identify attention dispersion as the underlying...

https://arxiv.org/abs/2603.14184

56 CONF

How Open Must Language Models be to Enable Reliable Scientific Inference?

ArXiv CS.AI 50%

arXiv:2603.26539v2 Announce Type: replace-cross Abstract: How does the extent to which a model is open or closed impact the scientific inferences that can be drawn from research that involves it? In this paper, we analyze how restrictions on information about model construction...

https://arxiv.org/abs/2603.26539

57 CONF

AI-Powered Facial Mask Removal Is Not Suitable For Identification

ArXiv CS.AI 50%

arXiv:2603.27747v2 Announce Type: replace-cross Abstract: Recently, crowd-sourced online criminal investigations have used generative-AI to enhance low-quality visual evidence. In one high-profile case, social-media users circulated an "AI-unmasked" image of a federal agent...

https://arxiv.org/abs/2603.27747

58 CONF

Transcription and Recognition of Italian Parliamentary Speeches Using Vision-Language Models

ArXiv CS.AI 50%

arXiv:2603.28103v2 Announce Type: replace-cross Abstract: Parliamentary proceedings represent a rich yet challenging resource for computational analysis, particularly when preserved only as scanned historical documents. Existing efforts to transcribe Italian parliamentary...

https://arxiv.org/abs/2603.28103

59 CONF

Compute Aligned Training: Optimizing for Test Time Inference

ArXiv CS.AI 50%

arXiv:2604.24957v2 Announce Type: replace-cross Abstract: Scaling test-time compute has emerged as a powerful mechanism for enhancing Large Language Model (LLM) performance. However, standard post-training paradigms, Supervised Fine-Tuning (SFT) and Reinforcement Learning (RL),...

https://arxiv.org/abs/2604.24957

60 CONF

JoyAI-Image: Awaking Spatial Intelligence in Unified Multimodal Understanding and Generation

ArXiv CS.AI 50%

arXiv:2605.04128v2 Announce Type: replace-cross Abstract: We present JoyAI-Image, a unified multimodal foundation model for visual understanding, text-to-image generation, and instruction-guided image editing. JoyAI-Image couples a spatially enhanced Multimodal Large Language...

https://arxiv.org/abs/2605.04128

61 CONF

PBT-Bench: Benchmarking AI Agents on Property-Based Testing

ArXiv CS.AI 50%

arXiv:2605.15229v2 Announce Type: replace-cross Abstract: Existing code benchmarks measure whether an agent can produce any test that reproduces a known bug, or whether it can produce a patch that fixes a described issue. Neither isolates the distinct skill of property-based...

https://arxiv.org/abs/2605.15229